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# Automated RNAscope™ Multiplex Fluorescent In Situ Hybridization on the Parhelia Spatial Station

*Optimized protocol for automated single-molecule RNA detection with Opal TSA fluorophores in FFPE tissues*

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**Abstract**

The RNAscope™ Multiplex Fluorescent v2 Assay (Advanced Cell Diagnostics / Bio-Techne) enables simultaneous visualization of multiple RNA targets at single-molecule resolution in tissue sections using branched DNA signal amplification and tyramide signal amplification (TSA) chemistry with Opal fluorophores. However, manual execution of the multi-step protocol is labor-intensive, requires precise temperature control, and introduces operator-dependent variability. Here, we describe the successful adaptation and optimization of the RNAscope Multiplex Fluorescent v2 protocol for fully automated execution on the Parhelia Spatial Station. Using human tonsil and melanoma FFPE tissues, we systematically evaluated the effects of antigen retrieval buffer pH, peroxide pretreatment, wash stringency, and HRP blocking conditions on signal specificity and sensitivity. Our optimized protocol achieves highly specific co-detection of up to three RNA targets with excellent signal-to-noise ratios and is compatible with downstream protein detection via CODEX. The Parhelia Spatial Station's capillary gap technology, programmable coverpad lift, and precision temperature control address key challenges in automating the RNAscope workflow, including reagent uniformity, bubble prevention, and wash stringency control.

## Introduction

Spatial transcriptomics has emerged as a transformative approach for characterizing gene expression within the native tissue architecture. The RNAscope™ Multiplex Fluorescent v2 Assay, developed by Advanced Cell Diagnostics (ACD, a Bio-Techne brand), employs a proprietary branched DNA (bDNA) signal amplification method coupled with tyramide signal amplification (TSA) to visualize individual RNA transcripts as discrete fluorescent puncta within intact tissue sections. The assay uses target-specific double-Z probe pairs that bind adjacent sequences on the target mRNA, providing high specificity through the requirement for two independent binding events. Sequential signal amplification steps (AMP 1–3) build a branched amplification tree on each probe pair, followed by HRP-mediated covalent deposition of tyramide-conjugated Opal fluorophores, enabling multiplex detection of up to three or four RNA targets per tissue section.

Despite its powerful capabilities, the manual RNAscope protocol presents significant logistical challenges. The complete workflow spans approximately 12 hours and requires precise temperature control at 40°C for hybridization and amplification steps, rapid transitions between 40°C incubations and room-temperature washes, and careful attention to preventing tissue sections from drying out. These requirements make the protocol particularly susceptible to operator-dependent variability and limit throughput.

The Parhelia Spatial Station is an open-chemistry robotic staining platform based on capillary gap reagent exchange technology (Figure 1). Its sealed, humidity-controlled chamber and precision temperature control (0–100°C) make it well- suited for automating temperature-sensitive, multi-step spatial biology workflows. The capillary gap design enables uniform reagent distribution with minimal reagent consumption (approximately 120 µL per slide per step, compared to 300–400 µL required for manual protocols). Here, we describe the adaptation, optimization, and validation of the RNAscope Multiplex Fluorescent v2 protocol for automated execution on the Parhelia Spatial Station.

## Materials and Methods

### *Tissue Specimens*

Human tonsil FFPE sections were used for protocol optimization, providing a well-characterized lymphoid tissue with defined immune cell populations. Human melanoma FFPE sections and human Covid patients lung sections were used for validation experiments. All tissues were

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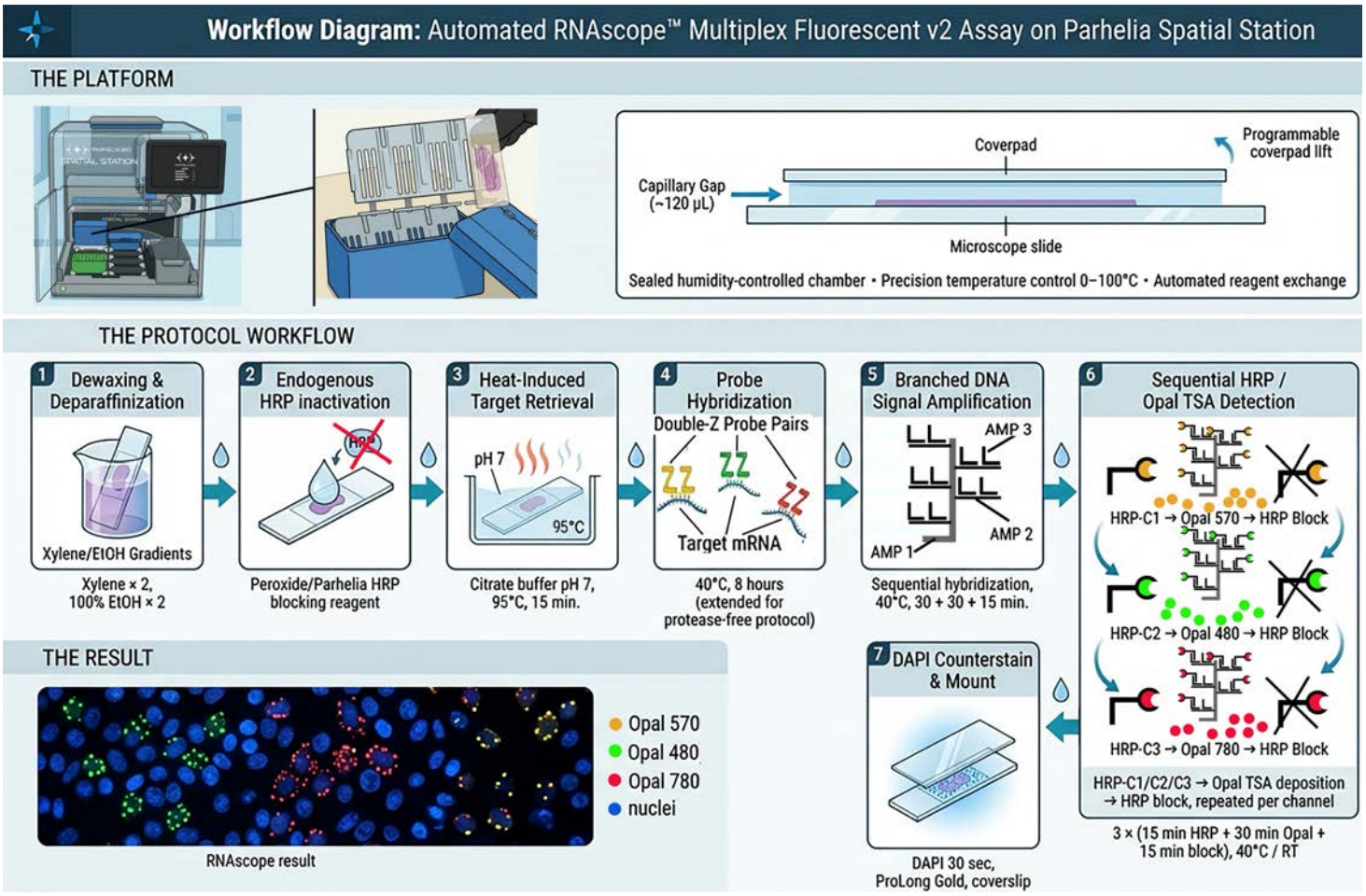


**Figure 1.** *Schematic overview of the automated RNAscope Multiplex Fluorescent v2 Assay workflow on the Parhelia Spatial Station. FFPE tissue sections undergo onboard dewaxing, heat-induced antigen retrieval, and hydrogen peroxide quenching, followed by probe hybridization, branched DNA signal amplification (AMP 1–3), and sequential HRP-mediated Opal fluorophore deposition with HRP blocking between channels. Counterstaining with DAPI completes the workflow.*

sectioned at 5 µm and mounted on SuperFrost Plus slides (Fisher Scientific).

### *RNAscope Probes and Reagents*

For tonsil optimization, CD163 (Channel 1) and CD8A (Channel 2) target probes were selected for their mutually exclusive expression pattern in macrophages and CD8+ T cells, respectively. For melanoma validation, CXCL9 (Opal 570), CCL19 (Opal 480), and CXCL10 (Opal 780) probes were used. All RNAscope Multiplex Fluorescent Reagent Kit v2 components (AMP 1–3, HRP-C1/C2/C3, HRP blocker, DAPI) were from ACD/Bio-Techne (Cat. No. 323100). Opal fluorophores (520, 570, 690, and Polaris 780) were from Quanterix (formerly Akoya Biosciences).

### *Automated Staining on the Parhelia Spatial Station*

All staining was performed on the Parhelia Spatial Station using the ST12 Slide Staining Module with capillary gap exchange. Protocols were designed and exported via Parhelia StainWorks software. The automated workflow included: onboard dewaxing using Parhelia Dewax reagent; heat-induced antigen retrieval at 95°C; hydrogen peroxide quenching; probe hybridization at 40°C for 8 hours; sequential amplification steps (AMP 1–3) at 40°C; channel-specific HRP incubation, Opal fluorophore deposition, and HRP blocking; and DAPI nuclear counterstaining.

### *Key Modifications for Automated Execution*

Several critical modifications were introduced for the automated platform. (1) **Protease treatment was omitted** to enable downstream co-detection of RNA with proteins. Hybridization time was extended from 2 to 8 hours to

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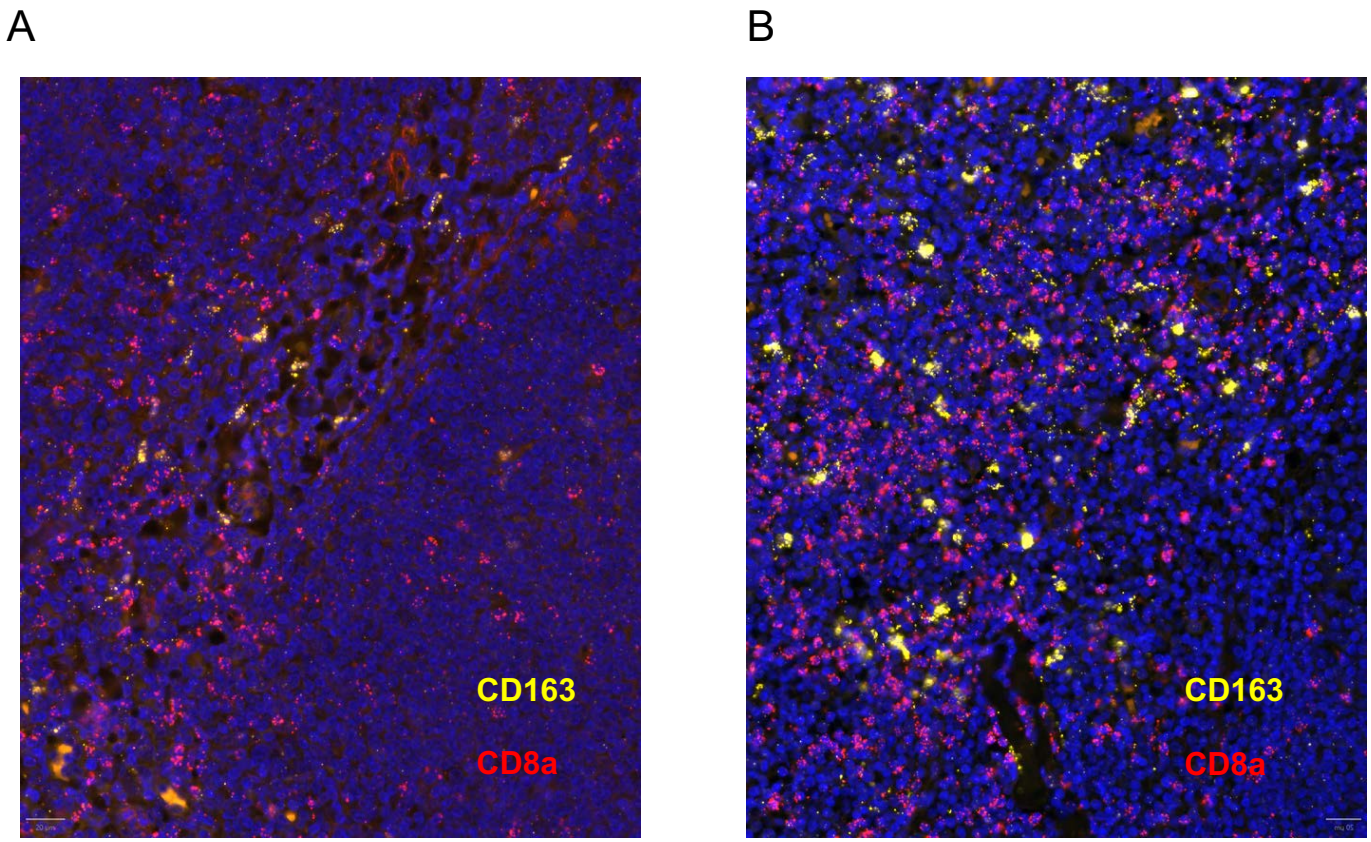


**Figure 2.** *RNAscope duplex detection of CD163 (Cy3) and CD8A (Cy5) in human tonsil FFPE tissue using pH 9 (A) and pH 7 (B) Note elevated tissue autofluorescence and reduced signal intensity compared to pH 7 conditions. DAPI (blue). Scale bar: 20 µm.*

compensate for the absence of protease digestion. (2) The standard RNAscope wash buffer was substituted for **Bond Wash reagent**. The manual protocol relies on rapid transitions from 40°C hybridization to room-temperature washes for stringent washing; because such transitions are impractical on a robotic platform, a higher-salt wash buffer (Bond Wash) was adopted for washes performed at hybridization temperature, with an additional high-salt storage buffer for transitions to room-temperature TSA steps. (3) **Programmable coverpad lift** cycles were introduced during prolonged incubations to remove air bubbles that formed within the capillary gap. (4) A **custom HRP blocking solution** (Parhelia) replaced the manufacturer-provided reagent to reduce background in multiplexed detection.

### *Optimization Parameters*

The following parameters were systematically evaluated on tonsil tissue: antigen retrieval buffer pH (pH 7 citrate vs. pH 9 alkaline); hydrogen peroxide pretreatment (included vs. omitted); wash duration and stringency; and HRP blocking reagent formulation. Signal specificity was assessed by examining the mutually exclusive expression of CD163 (macrophages) and CD8A (CD8+ T cells).

### *Imaging*

Stained slides were imaged on a PhenoCycler Fusion (PCF) fluorescence microscope (Quanterix, formerly Akoya)) equipped with DAPI, FITC, Cy3, Cy5, and Cy7 filter sets at 20–40X magnification. Composite images were generated to visualize marker co-expression and spatial distribution.

## Results

### *Antigen Retrieval Buffer pH Significantly Affects Signal Quality*

Comparison of citrate-based pH 7 and alkaline pH 9 antigen retrieval buffers revealed that pH 7 retrieval consistently produced superior signal intensity and specificity for RNAscope probes in tonsil tissue. **Figure 2 (B)** shows the pH 7 condition with clear, punctate RNAscope signals for both CD163 (Cy3, red) and CD8A (Cy5, magenta) against a clean DAPI (blue) nuclear background. In contrast, **Figure**

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**2 (A)** shows the pH 9 condition, which exhibited markedly higher background autofluorescence and reduced signal specificity, with brown-tinted tissue autofluorescence obscuring the specific RNAscope signals. Based on these results, pH 7 citrate retrieval buffer was adopted for all subsequent experiments.

### *Hydrogen Peroxide Pretreatment and HRP Blocking Optimization*

A key challenge encountered during protocol adaptation was frequent occurrence of non-specific signal, manifested as one to several fluorescent spots per cell that obscured quantification of low-abundance transcripts. Systematic investigation revealed two principal sources of this background. First, hydrogen peroxide pretreatment, recommended in the standard protocol for quenching endogenous peroxidase activity, was found to paradoxically increase non-specific signal. We hypothesize that peroxide treatment increases the number of positively charged sites in the tissue, promoting non-specific hybridization of negatively charged DNA probes. Omitting the peroxide pretreatment step yielded the best signal-to-noise ratios in tissues with low endogenous peroxidase levels, such as tonsil, and was essential in tissues with high myeloperoxidase content, such as immune-infiltrated lung.

Second, in multiplex experiments involving more than one target, background signal was primarily associated not with the peroxide pretreatment step but with the manufacturer-provided HRP blocking solution used between channel-specific cycles of HRP-mediated fluorophore deposition. Replacement with a custom Parhelia HRP blocking solution significantly reduced inter-channel background, enabling cleaner multiplexed detection.

### *Optimized Protocol Achieves Highly Specific Duplex RNA Detection*

The final optimized protocol, incorporating pH 7 retrieval, extended wash duration, custom HRP blocking, and programmable coverpad lift, produced highly specific co-detection of CD163 and CD8A in human tonsil tissue (**Figure 3**). CD163 expression (Cy3, yellow puncta) was confined to macrophages, while CD8A expression (Cy5, magenta puncta) was restricted to CD8+ T cells, demonstrating the expected mutually exclusive pattern with minimal cross-reactivity or non-specific background.

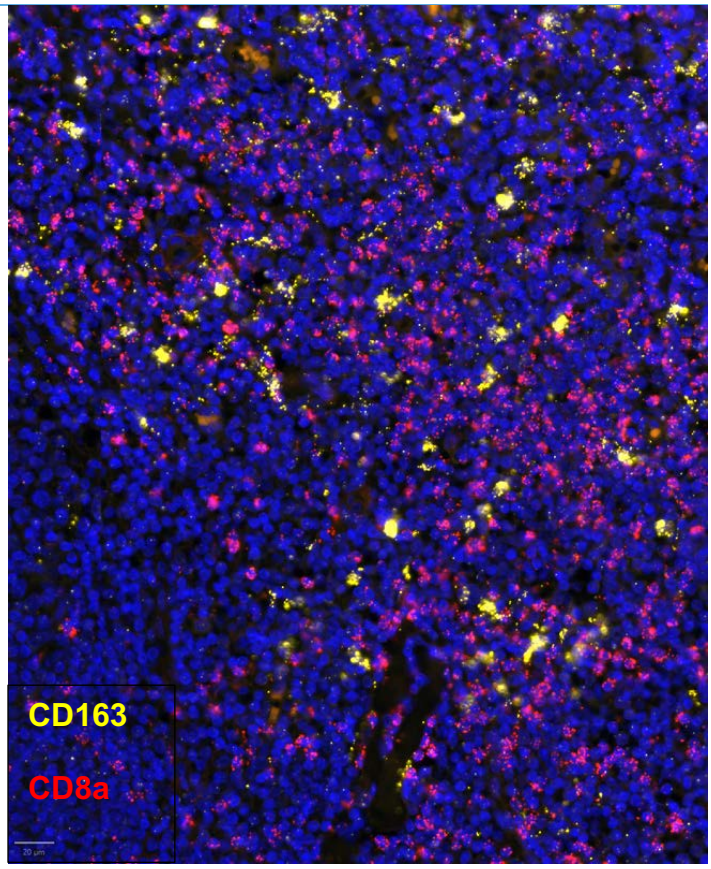


**Figure 3.** *Optimized automated RNAscope duplex detection of CD163 (Cy3, yellow) and CD8A (Cy5, magenta) in human tonsil FFPE tissue on the Parhelia Spatial Station. pH 7 retrieval, extended wash, custom HRP blocking, programmable coverpad lift. Note mutually exclusive expression patterns and minimal non-specific background. DAPI (blue). Scale bar: 20 µm.*

### *Validation: Multiplex Cytokine RNA Detection in Human Melanoma*

The optimized automated protocol was applied to human melanoma FFPE tissues for co-detection of three cytokine mRNAs: CXCL9 (Opal 570, red), CCL19 (Opal 480, green), and CXCL10 (Opal 780, yellow). The 2-plex (**Figure 4**) and 3-plex (**Figure 5**) results demonstrate clear, channel-specific signal with distinct spatial patterns. CXCL9 and CXCL10 expression was confined to myeloid cells (macrophages and dendritic cells) within the tumor microenvironment, whereas CCL19 was predominantly expressed in lymphatic endothelial cells and fibroblastic reticular cells (FRCs). These spatial expression patterns are consistent with the known biology of these chemokines in anti-tumor immune responses.

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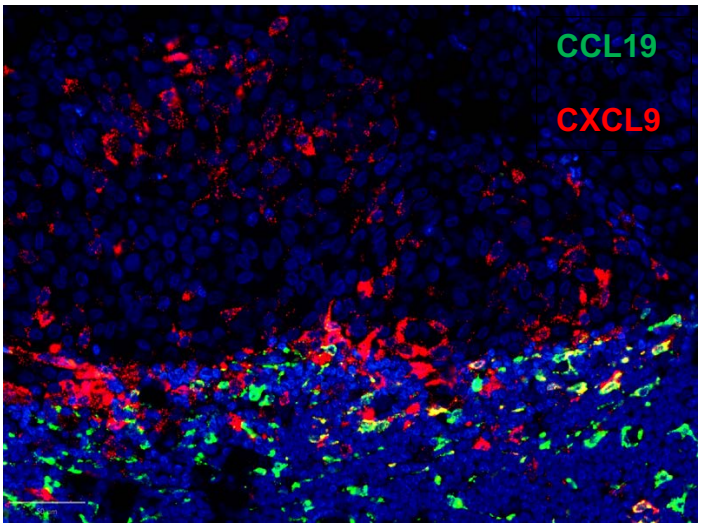


**Figure 4.** *Automated RNAscope 2-plex detection of CXCL9 (Opal 570, red) and CCL19 (Opal 480, green) mRNA in human melanoma FFPE tissue on the Parhelia Spatial Station. DAPI nuclear counterstain (blue).*

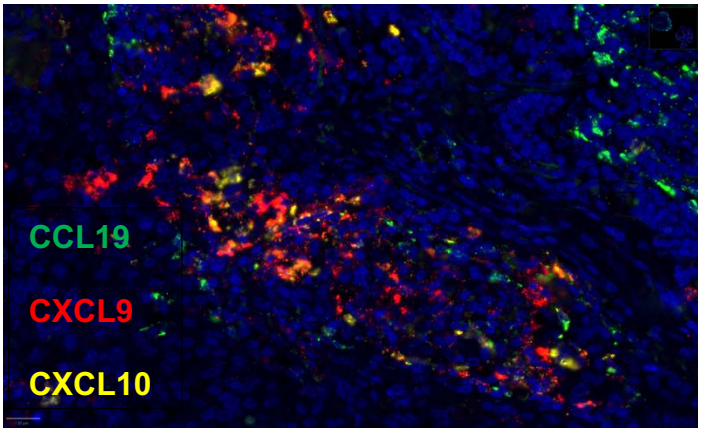


**Figure 5.** *Automated RNAscope 3-plex detection of CXCL9 (Opal 570, red), CCL19 (Opal 480, green), and CXCL10 (Opal 780, yellow) mRNA in human melanoma FFPE tissue. Note the distinct spatial localization of each cytokine transcript within the tumor microenvironment. DAPI (blue).*

***Compatibility with Downstream Protein Co-detection***

By omitting protease treatment from the RNAscope protocol, tissue antigenicity was preserved, enabling subsequent protein detection. Using the Parhelia Spatial Station and the protease-free RNAscope protocol, we achieved successful automated co-detection of RNA and proteins by following the RNAscope RNA detection workflow with CODEX (CO-Detection by indEXing) multiplexed immunofluorescence. This combined approach allowed co-detection of CXCL10 mRNA expression in lung tissue cells together with a highly multiplexed panel of immune and epithelial protein markers, demonstrating the potential for integrated RNA–protein spatial profiling on a single automated platform (**Figure 6**).

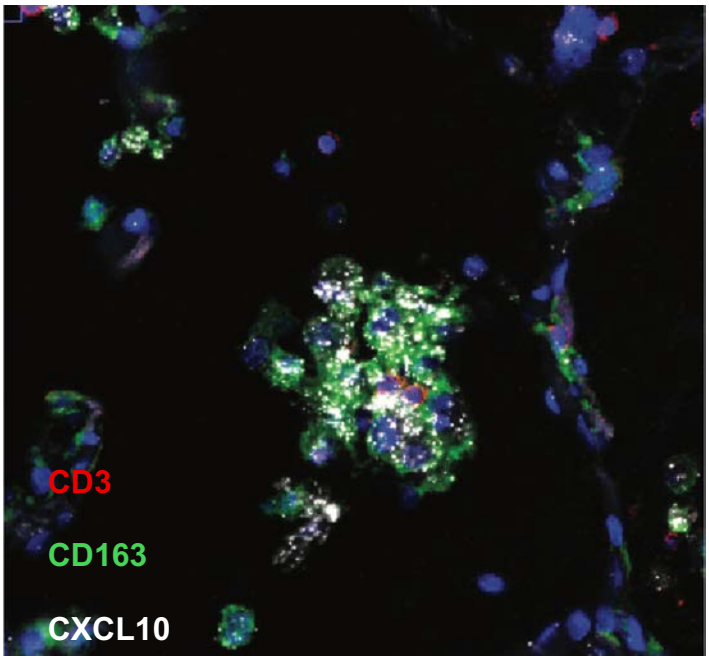


**Figure 6.** *Automated co-detection of proteins following the RNAscope detection of CXCL10 (Ax 647 tyramide, white), in human COVID lung on the Parhelia Spatial Station. DAPI nuclear counterstain (blue).*

## Discussion

The results presented here demonstrate that the Parhelia Spatial Station provides a robust platform for fully automating the RNAscope Multiplex Fluorescent v2 Assay on FFPE tissues. Systematic optimization of retrieval buffer pH, peroxide treatment, wash stringency, and HRP blocking conditions yielded highly specific multiplex RNA detection with excellent signal-to-noise ratios.

The finding that pH 7 citrate buffer outperformed pH 9 alkaline buffer for antigen retrieval in the RNAscope context is noteworthy. While alkaline retrieval is often preferred for protein-based immunohistochemistry, the lower pH condition appears to better preserve RNA accessibility while minimizing tissue autofluorescence. This is consistent with previous observations that high-pH treatment can increase non-specific binding of nucleic acid probes to tissue components.

The observation that hydrogen peroxide pretreatment can increase non-specific signal in RNAscope assays has practical implications for protocol design. While peroxide quenching is essential for tissues with high endogenous peroxidase activity (such as immune-infiltrated lung), it

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should be applied judiciously and may be omitted entirely in tissues with low peroxidase levels. Our hypothesis that peroxide treatment increases positively charged tissue sites, thereby promoting non-specific binding of negatively charged DNA probes, warrants further investigation.

| Parameter | Optimized Condition |
|---|---|
| **Antigen Retrieval** | pH 7 citrate buffer, 95°C |
| **Protease Treatment** | Omitted (extended hybridization compensates) |
| **Probe Hybridization** | 40°C, 8 hours |
| **Wash Buffer** | Bond Wash (higher salt) at hybridization temperature |
| **HRP Blocking** | Custom Parhelia HRP blocking solution |
| **Coverpad Lift** | Programmable periodic lift during prolonged incubations |
| **$H_2O_2$ Pretreatment** | Omitted for tissues with low endogenous peroxidase |

**Table 1.** *Summary of key optimization parameters for automated RNAscope on the Parhelia Spatial Station.*

The replacement of the manufacturer-provided HRP blocking solution with a custom Parhelia formulation was critical for achieving clean multiplex results. In sequential HRP-mediated fluorophore deposition, incomplete blocking of HRP activity between channels leads to non-specific deposition of subsequent fluorophores, generating false co-localization signals. The custom blocking solution more effectively inactivated residual HRP activity, enabling sequential channel development with minimal inter-channel bleed.

The Parhelia Spatial Station's unique engineering features were instrumental in overcoming several challenges specific to automating the RNAscope workflow. The capillary gap design ensured uniform reagent distribution with economical reagent usage; the programmable coverpad lift addressed bubble formation during extended incubations; and the precision temperature control enabled the critical wash stringency adjustments required when substituting Bond Wash for standard RNAscope wash buffer. The adoption of higher-salt washes at hybridization temperature, rather than the manual protocol's approach of rapid cooling to room temperature, represents a practical solution to the thermal transition challenge inherent in robotic staining platforms.

The successful co-detection of RNA and proteins on the same tissue section by combining the protease-free RNAscope protocol with CODEX immunofluorescence opens significant possibilities for integrated spatial profiling. By preserving tissue antigenicity through the omission of protease digestion, researchers can simultaneously interrogate transcriptomic and proteomic landscapes within the same cells, providing a more comprehensive view of cellular states and microenvironmental interactions.

## Conclusions

We have demonstrated successful automation of the RNAscope Multiplex Fluorescent v2 Assay on the Parhelia Spatial Station for FFPE tissue analysis. Key findings include (**Table 1**): (1) pH 7 citrate antigen retrieval buffer provides superior signal quality compared to pH 9 alkaline buffer; (2) hydrogen peroxide pretreatment can increase non-specific probe binding and should be used judiciously; (3) a custom Parhelia HRP blocking solution significantly reduces inter-channel background in multiplex experiments; (4) extended probe hybridization (8 hours) compensates for omission of protease treatment, enabling downstream protein co-detection; and (5) the combination of Bond Wash buffer, programmable coverpad lift, and precision temperature control addresses the unique challenges of robotic RNAscope execution. The validated protocol achieves highly specific co-detection of up to three RNA targets and is compatible with subsequent protein detection via CODEX, supporting integrated RNA–protein spatial profiling on a single automated platform.

## Acknowledgments

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